

Certified Fuzzy Control by Least Action

An H^1 Variational Formulation for TSK Systems, with
Sum-of-Squares Stability Certificates and a Calibrated
Performance/Certificate Trade

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Abstract

We develop a variational formulation of Takagi–Sugeno–Kang (TSK) fuzzy inference in which model fitting is posed as minimization of an H^1 (Sobolev) action functional, and we carry the formulation through to closed-loop control with machine-checked stability certificates. Four results are analytic. First, with antecedents fixed the consequent block is the *global* minimizer of the action in closed form, so the only non-convexity in the problem is the antecedent placement; variable projection then yields an exact reduced gradient via the envelope theorem. Second, the natural design question—when does sequential (greedy) rule identification coincide with the joint fit—has the exact answer “iff the H^1 Gram matrix is block-diagonal,” and for two equal-width Gaussians the slope weight that achieves H^1 orthogonality has the closed form $\lambda^* = 3b^4/(8c^2)$, equivalently $\ell^* = \sqrt{6}w$ where w is the crossover width. A factor-of-four obstruction shows no single λ can orthogonalize three or more uniformly spaced rules. Third, on the control side the closed-loop suboptimality gap is *exactly* an integral of a Bregman divergence for any convex control cost, not merely the quadratic case. Fourth, switching to π -shaped (rational) membership functions makes the closed loop a rational vector field, and a sum-of-squares S-procedure certificate for the region of attraction is then verified in exact rational arithmetic, with no floating-point step in the chain.

Empirically, on a two-mass-spring benchmark scored jointly on settling time, peak force and energy, the least-action fit converges in two rules (10 parameters, score 0.867 against an LQR baseline of 1.000) and then degrades through distribution shift rather than through any deficiency of the fit. Direct policy optimization warm-started from the least-action fit reaches 0.655, improving all three objectives simultaneously, while a cold start fails outright—the variational fit is the initializer that makes the optimizer reachable. Finally we make the stability certificate an *objective* rather than an afterthought. A sampled necessary condition for the sum-of-squares program, evaluated by ray search, estimates the certified region 2500 $\times$ faster than the semidefinite program with a residual bias of a single constant $\kappa = 0.962$; specifying a certified radius then yields a controller the solver certifies at or above it. The best controller so obtained carries a certified ball of 0.679 at score 0.847, dominating the imitation fit on both axes.

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1 Introduction

This paper develops an idea originating in a handwritten derivation by the second author: that fitting a fuzzy inference system should be posed as a principle of least action rather than as least squares, and that the resulting stationarity conditions should be strong enough to certify optimality rather than merely to terminate an optimizer. The motivation is that a TSK system is a smooth parametric model, and smooth parametric models admit variational treatment; what is unusual is insisting that every step of the treatment produce a *statement that can be checked*, rather than a number produced by a solver.

Two questions organize the work.

1. **Approximation.** Given a target field y_d on an input universe X , what is the right functional to minimize over TSK parameters, and what can be proved about its minimizers? In particular: under what conditions does sequential rule identification (fit rule 1, fit rule 2 to the residual, and so on) agree with the joint fit? This was the question that prompted the work.
2. **Control.** If the same machinery is turned on a control problem, what is provable about the resulting controller—not measured, proved? We ask for suboptimality certificates and for stability certificates, and we ask what happens when the two are placed in competition.

Contributions. The specific claims are:

- An H^1 action functional for TSK approximation whose consequent block has a closed-form global minimizer (Theorem 3), an exact analytic reduced gradient (Theorem 5), and second-order conditions stated correctly on the critical cone (§3.3).
- An exact characterization of when greedy identification equals the joint fit (Proposition 6), a closed form for the H^1 -orthogonalizing slope weight (Theorem 8), and a structural obstruction showing it cannot be achieved for $N \geq 3$ uniformly spaced rules (Corollary 9).
- The classifier-side results that make defuzzification differentiable: Mamdani aggregation equals summation iff output supports are disjoint (Theorem 10), centroid defuzzification of such a system is an order-0 TSK model (Theorem 11), and an explicit error bound for β -annealed mean-of-maxima (Theorem 12).
- An exact suboptimality certificate for arbitrary convex control cost, expressed as an integral of a Bregman divergence (Theorem 14); the familiar quadratic identity is a special case.
- A sum-of-squares region-of-attraction certificate for a π -membership TSK closed loop, verified in exact rational arithmetic (§8), together with the practical finding that the binding obstruction was solver accuracy and not the relaxation.
- A certificate-aware policy optimization in which the certified radius enters the objective (§9), and a calibration procedure that turns the certified radius from a dial into a specification (§10).

We are explicit throughout about what is proved, what is certified numerically, and what is merely measured; §11 tabulates this.

2 Model and action functional

2.1 The model

On a scalar input universe $X = [x_{lo}, x_{hi}]$, an order- p TSK system with N rules is

$$y_c(x) = \sum_{i=1}^N \varphi_i(x) f_i(x), \quad \varphi_i = \frac{\mu_i}{\sum_j \mu_j}, \quad f_i(x) = \sum_{k=0}^p B_{ik} x^k, \quad (1)$$

where μ_i is the membership function of rule i with centre a_i and width b_i . The normalized firing strengths form a partition of unity, $\sum_i \varphi_i \equiv 1$. Writing $\psi_{ik} = \varphi_i x^k$ for the *rule regressors* and θ for the flattened coefficients B , the model is linear in θ :

$$y_c(x) = \theta^\top \psi(x). \quad (2)$$

This is the structural fact the entire paper exploits. The parameters split into a block in which the model is linear (θ) and a block in which it is not (a, b).

Remark 1. In implementation the monomial x^k is replaced by a rescaled monomial $t = (x - \text{mid})/\text{half}$. This spans the identical function space, so the model is unchanged, but it prevents the Gram condition number from reaching 10^7 on a universe such as $[-15, 15]$.

2.2 The action

Let $e = y_d - y_c$ be the error field. We take as action the squared H^1 norm of the error,

$$S[y_c] = \int_X \left(e^2 + \lambda (e')^2 \right) dx = \|e\|_{H^1}^2, \quad \lambda = \ell^2 \geq 0, \quad (3)$$

where ℓ has units of x and plays the role of a correlation length. The L^2 term penalizes point-wise error; the λ term penalizes error in slope, which is what makes the formulation a genuinely variational one rather than a reweighted regression.

Euler–Lagrange. Treating y_c as a free field, the first variation of (3) vanishes when

$$\lambda e'' - e = 0, \quad (4)$$

the screened-Poisson (modified Helmholtz) equation, with natural boundary conditions $e' = 0$ at the endpoints. Its only solution decaying in both directions is $e \equiv 0$: in the unconstrained function space, least action recovers the target exactly. The content of the method is therefore entirely in what happens when y_c is restricted to the TSK manifold (1).

Stationarity in the model class. Restricting to (2), $\delta S = 0$ in the θ directions is

$$\langle e, \psi_m \rangle_{H^1} = 0 \quad \text{for every regressor } \psi_m, \quad (5)$$

i.e. *Galerkin orthogonality*: the error is H^1 -orthogonal to the model subspace. Equation (5) is the weak form of (4) and is the correct stationarity condition for the finite-dimensional problem. It is what we check numerically, and it is what distinguishes this formulation from a generic nonlinear least-squares fit.

Remark 2 (Convexity). For $\lambda \geq 0$ the map $y_c \mapsto S$ is a strictly convex quadratic on function space, so the variational problem has no conjugate points and the Jacobi condition is vacuous. All non-convexity in the parametric problem enters through the map $(a, b) \mapsto \psi$, not through the action.

3 The consequent block is globally solvable

3.1 Closed form

Define the H^1 Gram matrix and right-hand side

$$G_{mn} = \langle \psi_m, \psi_n \rangle_{H^1}, \quad r_m = \langle y_d, \psi_m \rangle_{H^1}. \quad (6)$$

Theorem 3 (Variable projection). *For fixed antecedents (a, b) , the action (3) is a convex quadratic in θ with Hessian $2G \succeq 0$, and every minimizer satisfies the normal equations $G\theta = r$. If the regressors are linearly independent in H^1 then $G \succ 0$ and $\theta^* = G^{-1}r$ is the unique global minimizer. The reduced action is*

$$\hat{S}(a, b) = \|y_d\|_{H^1}^2 - r^\top G^{-1}r. \quad (7)$$

Proof. Substituting (2) into (3), $S = \|y_d\|_{H^1}^2 - 2\theta^\top r + \theta^\top G\theta$. The Hessian $2G$ is a Gram matrix of an inner product and hence positive semidefinite, so S is convex and stationarity is sufficient. Stationarity is $G\theta = r$, which is (5) written out. Substituting θ^* gives (7). $\square$

Theorem 3 is the strongest guarantee in the paper and it is free: no iteration, no initialization, no local minima. It also localizes the difficulty precisely—the problem is non-convex *only* in (a, b) .

Remark 4 (Regularization). Numerically we solve $(G + \varepsilon \text{tr}(G)/m I)\theta = r$ with $\varepsilon = 10^{-9}$. The ridge is trace-relative so it scales with the problem, and it is a ridge rather than a truncating pseudo-inverse deliberately: a rank-flipping SVD makes the map $(a, b) \mapsto \theta$ discontinuous, which destroys the differentiability the next section depends on.

3.2 Exact reduced gradient

Finite differences on (7) are unreliable, because $\hat{S}$ is computed only to accuracy $\kappa(G)\varepsilon_{\text{mach}}$; in our experiments SciPy’s default step produced a gradient dominated by round-off, and L-BFGS-B terminated after zero iterations reporting success. The envelope theorem removes the issue.

Theorem 5 (Analytic reduced gradient). *Let $\theta^*(a, b)$ be as in Theorem 3. Then for any antecedent parameter $p \in \{a_i, b_i\}$,*

$$\frac{\partial \hat{S}}{\partial p} = -2 \left\langle e, \frac{\partial y_c}{\partial p} \right\rangle_{H^1}, \quad \frac{\partial y_c}{\partial p} = \theta^{*\top} \frac{\partial \psi}{\partial p}, \quad (8)$$

with no contribution from $\partial \theta^/\partial p$.*

Proof. $\hat{S}(a, b) = S(\theta^*(a, b), a, b)$, so by the chain rule $\partial_p \hat{S} = \partial_\theta S \cdot \partial_p \theta^* + \partial_p S$. At θ^* the first factor vanishes by Theorem 3. The remaining partial is (8). $\square$

The derivative $\partial \psi / \partial p$ is available in closed form from the membership-function derivatives $(\mu, \partial_x \mu, \partial_a \mu, \partial_b \mu, \partial_{xa}^2 \mu, \partial_{xb}^2 \mu)$; the ∂_x terms are required because the H^1 inner product differentiates the regressors. Replacing finite differences by (8) was the difference between an optimizer that stalled immediately and one that converged.

3.3 Second-order conditions, stated correctly

Three errors are easy to make here and all three occurred during this work.

1. **Test the reduced Hessian, not the full one.** The full-parameter Hessian is never definite: consequent rescaling and rule relabeling guarantee flat and symmetric directions. Only $\hat{S}$ has a meaningful Hessian.
2. **Fitted solutions sit on identifiability constraints.** We impose a minimum rule gap ($0.6 \times \text{pitch}$) and width bounds ($[0.25, 1.5] \times \text{pitch}$) so that rules remain distinguishable. At an active constraint the correct second-order condition is definiteness on the *critical cone*—the null space of the active constraint normals—not on all of $\mathbb{R}^{2N}$; and the correct first-order condition is that the *projected* gradient vanishes, the raw gradient being balanced by KKT multipliers.
3. **Curvature is informative only at a stationary point.** Galerkin orthogonality (5) certifies the consequent block by construction and therefore says nothing about (a, b) ; the antecedent gradient must be checked separately.

Our certificate accordingly reports the KKT residual on the critical cone, the projected-Hessian spectrum, and a falsification probe that walks along the most-negative eigenvector with the first-order term subtracted, to test whether the action actually decreases. Because the Hessian is obtained by finite differences, the verdict is tolerance-based: we compare against an estimated noise floor ε_f/h^2 rather than against exact zero, since judging against zero reads numerical dust as a saddle. This is the one place where we certify *numerically* rather than analytically (guarantee G11 in §11).

4 Rule orthogonality and sequential identification

The original design question was whether rules can be constrained so that $\int f_i f_j dx = 0$ for $i \neq j$, and whether rules can then be identified sequentially from residuals. Both halves have exact answers, and they are not the expected ones.

4.1 Greedy equals joint iff the Gram is block-diagonal

Partition the regressor index set by rule, and let G be blocked accordingly.

Proposition 6. *Sequential fitting—fit rule 1, then fit rule 2 to the residual, and so on—returns the joint minimizer of (3) for every target y_d if and only if G is block-diagonal with respect to the rule partition.*

Proof. If G is block-diagonal the normal equations decouple into one system per rule and greedy solves each exactly. Conversely, suppose $G_{IJ} \neq 0$ for some $I \neq J$. Choose y_d in the span of block J 's regressors. Greedy assigns block I a nonzero coefficient (since $r_I = G_{IJ}\theta_J \neq 0$), whereas the joint solution assigns block I zero; the two differ. $\square$

Note the quantifier: block-diagonality is required for agreement on *all* targets. For a particular y_d greedy may coincide with the joint fit accidentally.

Proposition 7. *For $\lambda > 0$, block-diagonality of G does not require disjoint rule supports.*

Proof sketch. The block entry is $\int \varphi_i \varphi_j dx + \lambda \int \varphi'_i \varphi'_j dx$. For adjacent normalized rules the first integral is strictly positive and the second is strictly negative, because the partition of unity forces $\varphi'_i = -\varphi'_j$ in the two-rule crossover. Cancellation is therefore possible at a strictly positive λ with overlapping supports. $\square$

Proposition 7 corrects a claim we initially made in the other direction, obtained by setting $\lambda = 0$ in a statement whose hypothesis required $\lambda > 0$.

4.2 The orthogonalizing slope weight in closed form

Proposition 7 raises the constructive question: what value of λ achieves cancellation? For the two-rule case it is elementary.

Theorem 8 (λ^*). *For two equal-width Gaussian membership functions with centres $\pm c$ and width b on $\mathbb{R}$, the normalized weights collapse to a logistic, $\varphi_1 = \sigma(kx)$ with $k = 4c/b^2$, and*

$$\int_{\mathbb{R}} \varphi_0 \varphi_1 dx = \frac{1}{k}, \quad \int_{\mathbb{R}} \varphi'_0 \varphi'_1 dx = -\frac{k}{6}, \quad (9)$$

so that the unique λ making the two rules H^1 -orthogonal is

$$\boxed{\lambda^* = \frac{6}{k^2} = \frac{3b^4}{8c^2}} \iff \ell^* = \sqrt{\lambda^*} = \sqrt{6} w, \quad w = \frac{1}{k} = \frac{b^2}{4c}. \quad (10)$$

Proof. The log-ratio $\log(\mu_1/\mu_0) = 4cx/b^2$ is linear in x , hence $\varphi_1 = \sigma(kx)$ with $k = 4c/b^2$ and $\varphi_0 \varphi_1 = \sigma(1 - \sigma) = \sigma'(kx)$, $\varphi'_i = \pm k \sigma'$. Substituting $u = kx$ and then $s = \sigma(u)$, $ds = s(1 - s)du$: $\int \varphi_0 \varphi_1 dx = k^{-1} \int_{\mathbb{R}} \sigma'(u) du = k^{-1}$ and $\int \varphi'_0 \varphi'_1 dx = -k \int_{\mathbb{R}} \sigma'(u)^2 du = -k \int_0^1 s(1 - s) ds = -k/6$. Setting $k^{-1} - \lambda k/6 = 0$ gives the result. $\square$

Interpretation. The target y_d appears nowhere in Theorem 8. Therefore λ^* is *not* a distinguished correlation length of the data; it is the correlation length of the rule crossover region, fixed entirely by partition geometry. The ratio $\ell^*/w = \sqrt{6} \approx 2.44949$ was confirmed numerically in every configuration tested.

Corollary 9 (Why $N \geq 3$ fails). *On a uniform partition of pitch d , adjacent pairs (separation d) and next-nearest pairs (separation $2d$) demand orthogonalizing weights in the ratio $(2d/d)^2 = 4$, since $\lambda^* \propto c^{-2}$. A single scalar λ therefore cannot orthogonalize three or more uniformly spaced rules.*

The measured ratio was 4.0000 in every case, and this explains why the residual off-block coupling plateaus near 10^{-2} for $N \geq 3$ rather than approaching zero.

The price of λ^* . Choosing $\lambda = \lambda^*$ trades L^2 accuracy for decoupling. Table 1 measures that trade against the L^2 -optimal ($\lambda = 0$) fit of the same partition, target $\tanh(x/3)$.

The cost stays below 8% across the sweep and below 2% for sharp crossovers. The closed form explains why: $\ell^* = \sqrt{6} b^2/(4c)$ is sub-rule-scale for any sensible partition, and weighting slopes at a short correlation length barely perturbs the L^2 solution.

Table 1: Cost of imposing H^1 orthogonality, relative to the L^2 -optimal fit of the same partition.

c	b	w	λ^*	ℓ^*	L^2 at $\lambda=0$	L^2 at λ^*	cost
5	1	0.050	0.015	0.12	3.2540	3.2542	0.01%
5	2	0.200	0.240	0.49	1.0992	1.1036	0.40%
5	4	0.800	3.840	1.96	0.5191	0.5394	3.91%
5	6	1.800	19.44	4.41	0.1799	0.1938	7.72%
5	8	3.200	61.44	7.84	0.7769	0.8204	5.60%
3	4	1.333	10.67	3.27	0.1101	0.1174	6.59%

4.3 The practical alternative

Since exact orthogonality is unavailable for $N \geq 3$, the usable route is to orthogonalize the *regressors* rather than the rules: an H^1 Gram–Schmidt (orthogonal least squares) pass on ψ , selecting rules greedily by residual reduction. This restores the greedy/joint agreement of Proposition 6 by construction, at the cost of rules that are no longer individually interpretable in the original basis.

5 Classifier case: making defuzzification differentiable

Without recurrence, a classifier ranking is a regression model with range $[0, 1]$, so the same action applies. The obstruction is that the standard defuzzifications—De Baets’ first/last/mean of maxima—are discontinuous, and a discontinuous output has no action to minimize. Three results remove the obstruction.

Theorem 10 (Aggregation equals summation). *Let rules have output membership functions $\{\nu_i\}$ on Y . Then $\bigvee_i \alpha_i \nu_i = \sum_i \alpha_i \nu_i$ pointwise for all firing strengths α if and only if the supports of $\{\nu_i\}$ are pairwise disjoint.*

Proof. If supports are disjoint, at most one term is nonzero at each y , so max and sum agree. Conversely if $\nu_i(y)\nu_j(y) > 0$ for some y and $i \neq j$, take $\alpha_i = \alpha_j = 1$: the sum strictly exceeds the max. $\square$

Theorem 11 (Centroid is order-0 TSK). *Under the hypotheses of Theorem 10 with product implication, centroid defuzzification returns $\hat{y} = \sum_i \varphi_i \bar{y}_i$ where $\bar{y}_i$ is the centroid of ν_i and φ_i the normalized firing strength; i.e. exactly an order-0 TSK model, whose output range $[0, 1]$ is free rather than imposed.*

Theorem 12 (Annealed mean-of-maxima bound). *Let $\hat{y}_\beta$ be the softmax-annealed (β -weighted) defuzzification and $\hat{y}_\infty$ the exact mean-of-maxima. If D is the diameter of the output universe and $r < 1$ is the ratio of the second-largest to the largest firing strength, then*

$$|\hat{y}_\beta - \hat{y}_\infty| \leq \frac{(N-1) D r^\beta}{1 + (N-1)r^\beta}. \quad (11)$$

Theorem 12 makes the approximation quantitative: the annealed score is C^∞ in the input, and the gap to the crisp defuzzification is controlled explicitly by β and by how decisively the rules separate. Together, Theorems 10–12 deliver the continuity and differentiability the action requires, which was the motivating request.

6 Control: an exact suboptimality certificate

6.1 The degenerate case

Theorem 13 (TSK reproduces LQR exactly). *For an LTI plant with quadratic cost, an order-1 TSK controller with any partition of unity and consequents $f_i(x) = -Kx$ for all i satisfies $u(x) = \sum_i \varphi_i(x)f_i(x) = -Kx$, the LQR optimum, identically.*

The proof is $\sum_i \varphi_i \equiv 1$. We record Theorem 13 because it is the correct baseline and because it is *vacuous as a recommendation*: it says the model class contains the linear optimum, so any benchmark on which LQR is optimal cannot discriminate. This is why §7 scores on settling time, peak force and energy jointly—objectives under which LQR is not optimal—rather than on the quadratic cost it was designed to minimize.

6.2 The exact certificate, for any convex cost

Consider a control-affine plant $\dot{x} = f(x) + g(x)u$ with running cost $\ell(x, u) = q(x) + c(u)$, optimal value V^* and optimal policy u^* .

Theorem 14 (Bregman suboptimality identity). *Let c be convex and differentiable, let V^* be differentiable and satisfy the Hamilton–Jacobi–Bellman equation, and let $u(\cdot)$ be admissible (the closed loop converges to the origin with $V^*(x(t)) \rightarrow 0$). Then*

$$\boxed{J(x_0) - V^*(x_0) = \int_0^\infty D_c(u(t) \| u^*(x(t))) dt} \quad (12)$$

where $D_c(u \| v) = c(u) - c(v) - c'(v)(u - v)$ is the Bregman divergence of c .

Proof. HJB stationarity gives $c'(u^*) = -g^\top \nabla V^*$. Hence

$$\ell(x, u) + \nabla V^*(f + gu) = q + c(u) + \nabla V^*f + \nabla V^*gu = c(u) - c(u^*) - c'(u^*)(u - u^*) = D_c(u \| u^*),$$

using $q + \nabla V^*f + \nabla V^*gu^* + c(u^*) = 0$ (HJB). The left side is $\ell + \frac{d}{dt}V^*(x(t))$ along the closed loop; integrating from 0 to ∞ and using admissibility to kill the boundary term gives (12). $\square$

Corollary 15. *For $c(u) = u^\top Ru$, $D_c(u \| u^*) = (u - u^*)^\top R(u - u^*)$ and (12) reduces to the familiar $J - V^* = \int (u - u^*)^\top R(u - u^*) dt$.*

Three features deserve emphasis. The identity is *exact*—no constants, no inequalities. Non-negativity of the gap is free, because $D_c \geq 0$ is convexity of c . And it holds for costs that are not polynomial: we verified it numerically for $c(u) = \cosh u - 1$ and $c(u) = u^2 + u^4$ on inverse-optimal benchmarks, with residuals of 10^{-11} to 10^{-12} (Table 2).

The hypotheses bind. Admissibility is not decorative. A controller fitted without the constraint $u(0) = 0$ leaves a steady-state offset, the closed loop settles to a nearby point rather than the origin, $V^*(x(t)) \not\rightarrow 0$, and the identity silently fails. We encountered exactly this, and it is also the reason the constraint appears in §8.

What lies outside. Theorem 14 certifies integral functionals of the closed loop. Peak force $\max_t |u|$ and settling time are not integral functionals—one is a supremum, the other an exit time—so two of the three benchmark objectives in §7 are provably outside the certified envelope, and are reported as measurements.

Table 2: Numerical check of (12) on inverse-optimal benchmarks.

cost	controller	gap	$\int D_c dt$	residual
$\cosh u - 1$	$0.5u^*$	3.099e−1	3.099e−1	3.2e−12
$\cosh u - 1$	$1.7u^*$	5.172e−1	5.172e−1	4.0e−11
$\cosh u - 1$	$-0.4x$	7.536e−1	7.536e−1	2.8e−11
$u^2 + u^4$	$0.5u^*$	1.857133e−1	1.857133e−1	1.1e−11
$u^2 + u^4$	$-0.4x$	8.955264e−2	8.955264e−2	1.4e−11

Table 3: Convergence with rule count. “shift” is the maximum distance from a closed-loop state to the nearest training sample, in units of the training set’s per-axis spread.

rules	params	settle	peak $ u $	energy	score	shift	settled
LQR	4	19.797	1.0000	0.5375	1.0000	—	yes
open-loop ref	—	10.972	0.8615	0.5585	0.8183	—	yes
1	5	26.773	0.9575	0.4214	1.0313	3.05	yes
2	10	17.830	0.9088	0.4248	0.8666	3.07	yes
3	15	17.217	0.9023	0.4390	0.8629	3.05	yes
4	20	18.470	0.9088	0.4465	0.8908	3.07	yes
6	30	18.257	0.8560	0.4798	0.8903	3.07	yes
8	40	19.040	0.5531	0.6121	∞	3.07	no
12	60	14.313	1.0000	1.2041	1.3211	4.67	yes
16	80	34.990	1.0000	7.8944	∞	6.73	no

7 Benchmark: two-mass-spring

7.1 Setup

Two unit masses coupled by a unit spring, force applied to the first mass, saturating at $|u| \leq 1$; state $z = (x_1, x_2, v_1, v_2)$. Controllers are scored by the equal-weight normalized scalarization

$$\text{score} = \frac{1}{3} \left(\frac{T_{\text{settle}}}{T_{\text{ref}}} + \frac{\max_t |u|}{|u|_{\text{ref}}} + \frac{\int u^2 dt}{E_{\text{ref}}} \right), \quad (13)$$

with the reference the best LQR over a gain sweep, so LQR scores 1.000 by construction. The question asked was how fast the least-action fit converges *with no refinement*: the recipe is applied once, with no tuning—weighted k -means rule placement on open-loop optimal trajectories, then variable projection of the consequents (one linear solve, globally optimal by Theorem 3), then simulate.

7.2 It converges in two rules

Ten parameters take the score from 1.031 (worse than LQR) to 0.867; the third rule adds 0.4% and that is the end of it. Two rules close 75% of the gap between the best LQR and the open-loop reference, $(1.000 - 0.863)/(1.000 - 0.818)$. The win is specific rather than uniform: at $N = 3$ the fuzzy controller uses 18% less energy than the best LQR and 21% less than the open-loop reference it was trained on, and 10% less peak force, while being *worse* on settling time (17.2 against 11.0). It trades response speed for actuator economy.

Table 4: Direct policy optimization, warm-started from the least-action fit, 600 closed-loop evaluations each.

rules	params	imitation	direct	improvement
2	10	0.8666	0.6547	24.5%
3	15	0.8629	0.6729	22.0%
4	20	0.8908	0.7565	15.1%

7.3 Then it degrades, for a diagnosable reason

Past $N \approx 3$ the score degrades monotonically and by $N = 8$ the closed loop stops settling. This is not a numerical artifact and not a failure of the variable projection, which remains a globally optimal linear solve at every rule count. It is **distribution shift**, and the shift column of Table 3 tracks the failure exactly: 3.05 while the method works, 4.67 at $N = 12$, 6.73 at $N = 16$. The controller is fitted only on optimal trajectories, so it has no data off them; more rules mean each rule is supported by fewer samples and extrapolates harder. Hence:

Rule count and closed-loop quality are not monotonically related. The least-action fit is globally optimal *for the data it was given*, and the data is the binding constraint—not the model class and not the optimizer.

Off-trajectory augmentation (perturbed tubes, or one DAGger round) removes the collapse but does not move the plateau near 0.86; nor do higher-order consequents, which shift it by 1.6% while fit accuracy and closed-loop score become *inversely* related. Both experiments point at the objective rather than the model.

7.4 Direct policy optimization breaks the plateau

Testing the implied fix—stop fitting sampled targets, optimize the closed-loop objective directly over the FIS parameters—requires surrendering Theorem 3: the closed-loop objective is not quadratic in θ , so only derivative-free search remains. We use Powell on a shaped surrogate in which each discontinuous term is replaced by a finite analogue (soft time outside a tolerance ball; log-sum-exp peak; unchanged energy; plus a terminal-norm term that makes a non-settling controller *improvable* rather than merely rejected).

At two rules the optimized controller improves *all three objectives at once*—31% faster settling, 35% less peak force and 37% less energy than the best LQR—which no imitation controller did. It is 34.5% better than the best LQR, 22.9% better than the best imitation result, and 20% better than the open-loop reference.

The warm start is load-bearing. The same structure started from zero consequents instead of the least-action fit failed outright: 601 evaluations, no stabilizing controller found. The two stages are therefore complementary rather than competing.

The least-action fit is not the final method—it is the initializer that makes the final method reachable. On its own it caps out at 0.863; without it, direct optimization does not start.

This rehabilitates variable projection in a specific role: its value is not that it produces the best controller, but that it produces a *stabilizing* one in a single globally optimal linear solve, with no search and no initialization problem of its own.

Table 5: Gram minimum eigenvalue by solver and multiplier degree ($\checkmark$ = certified).

solver	σ degree	$\rho = 0.5$	$\rho = 0.8$	$\rho = 1.0$
SCS (first-order)	4	-0.158	-0.236	-0.250
CLARABEL (interior-point)	2	-1.7e-10 $\checkmark$	-1.6e-10 $\checkmark$	-4.9e-11 $\checkmark$
CLARABEL (interior-point)	4	-8.2e-10 $\checkmark$	-1.1e-09 $\checkmark$	-1.8e-10 $\checkmark$

8 Stability certification by sum of squares

8.1 Rational membership functions

Replacing Gaussians with π -shaped (Cauchy) membership functions $\mu_i(z) = (1 + \|(z - a_i)/b_i\|^2)^{-1}$ makes every μ_i rational, hence φ_i rational, hence the TSK control law exactly a ratio of polynomials $u(z) = P(z)/Q(z)$ with $Q > 0$ structurally. The closed loop is then a rational vector field and the Lyapunov decrease condition becomes a polynomial inequality after clearing denominators—which is what a sum-of-squares (SOS) program can decide.

8.2 The certificate

With $V(z) = z^\top Pz$, we certify $\dot{V} < 0$ on the sublevel set $\{V \leq \rho\} \setminus \{0\}$ via the S-procedure: find σ SOS with

$$W(z) - \varepsilon \|z\|^2 - \sigma(z)(\rho - V(z)) \text{ is SOS,} \quad (14)$$

where $W = -\dot{V} \cdot Q^2$ is the numerator-cleared decrease condition. Two implementation facts were decisive.

The constraint $u(0) = 0$ is not cosmetic. An unconstrained fit gives $u(0) = O(10^{-4})$, so the origin is *not* an equilibrium, the closed loop settles to a nearby offset point, and no Lyapunov function can certify asymptotic stability to a non-equilibrium. The same defect voids Theorem 14 through its admissibility hypothesis. Because $u(0) = 0$ is a single linear equality in θ , it is imposed exactly by a KKT system, so the fit remains one linear solve and remains the *global* optimum subject to the constraint—Theorem 3 survives intact.

Both SOS bases must start at degree 1. At $z = 0$ the target of (14) is $-\sigma(0)\rho < 0$, which cannot be SOS unless $\sigma(0) = 0$. Left unconstrained, the solver silently drives $\sigma(0) \rightarrow 0$ and every Gram matrix sits on the positive-semidefinite boundary at 10^{-10} . Forcing both bases to begin at degree 1 gives a genuine margin of 3.5×10^{-3} at the same certified ρ .

8.3 The obstruction was the solver

CLARABEL closes at multiplier degree 2—the degree at which SCS reported a violation of -0.25 —in well under a second. The diagnostic that identified this was the *relative* violation $|\lambda_{\min}|/\max|G|$: SCS bottoms out at 10^{-6} to 10^{-5} , a first-order method’s realistic accuracy on this problem, while CLARABEL reaches 10^{-11} . Reading the absolute eigenvalue suggested gross infeasibility; the relative figure showed it was noise. We record this because it cost four hours and because the conclusion generalizes: on small, badly scaled SOS programs, interior-point and first-order solvers are not interchangeable.

The certified region for the two-rule imitation controller is $\{z^\top Pz \leq 1.1144\}$.

8.4 Exact rational verification

An interior-point SDP certifies only to solver tolerance. We upgrade the result by the standard rounding-and-projection procedure (Peyrl and Parrilo): solve with a strictly positive margin; round both Gram matrices to rationals with bounded denominator; rebuild the target polynomial exactly over $\mathbb{Q}$ (IEEE doubles *are* rationals, so only the deliberate rounding introduces error); project the rounded Gram onto the affine set where the polynomial identity holds exactly, which decouples monomial-by-monomial and is available in closed form; and verify positive definiteness in exact arithmetic.

Two practical notes. Rational $LDL^\top$ is exact but useless at this size, because denominators square at each elimination step and the 14×14 factorization does not finish; integer leading principal minors computed by fraction-free (Bareiss) elimination have no such blow-up. And `nsimplify(..., rational=True)` must be avoided—it searches for “nice” closed forms and returns objects such as $2^{818/971}3^{651/971}$ for an ordinary double—in favour of the exact dyadic conversion.

The result: at $\rho = 1/2$ the projected identity has residual exactly 0, and all 14 leading principal minors of the 14×14 Gram are positive integers, the largest a 27,737-bit integer. Therefore $z^\top Pz$ decreases strictly on $\{z^\top Pz \leq 1/2\} \setminus \{0\}$ for the rationalized controller, proved rather than measured, with no floating-point step remaining in the chain.

8.5 Certification does not scale in rule count

Since $\deg Q = 2(N - 1)$, the SDP grows quickly: the Gram is 35×35 at $N = 2$, 70×70 at $N = 3$, 126×126 at $N = 4$ and 715×715 at $N = 8$. $N = 2$ solves in seconds; $N = 8$ is out of reach with these solvers. Certification is tractable only at low rule count. It happens to land well here, since §7 found convergence at two or three rules, but any future move toward more rules trades certifiability for performance.

9 Certificate-aware policy optimization

9.1 An uncontrolled trade

Certifying the directly optimized controller of §7 exposed a trade that nothing in the objective knew about: performance improved ($1.044 \rightarrow 0.773$) while the certified ball *shrank* ($0.568 \rightarrow 0.274$). The optimizer buys performance by acting more aggressively, and aggression is exactly what a Lyapunov certificate charges for.

9.2 A proxy that is safe to optimize against

Putting the certified radius into the objective is obstructed by cost: one SOS solve is ~ 1.5 s and a bisection over two Lyapunov candidates is 47 s per controller, against a 601-evaluation search. We therefore optimize a sampled surrogate,

$$\hat{\rho} = \min\{z^\top Pz : \dot{V}(z) \geq 0\}, \quad \hat{r} = \sqrt{\hat{\rho}/\lambda_{\min}(P)}, \quad (15)$$

estimated over a fixed sample set. Its logical status is what makes this sound: a state with $\dot{V}(z) \geq 0$ *proves* the SOS program must fail at that ρ , since a certificate is a statement about every point of the sublevel set. Hence $\hat{\rho}$ is an *upper bound* on any achievable certified ρ —a necessary condition, never sufficient. That is the right direction for a search objective and the wrong direction for a claim, so the proxy is optimized against and never reported: every controller is subsequently passed through the real semidefinite program.

Table 6: Certificate-aware policy optimization. “SOS ball” is the real certificate, best of two Lyapunov candidates; “held out” is the score on six initial conditions not used in the objective.

w	score	held out	proxy $\hat{r}$	SOS ball	V from
— (imitation)	1.0437	1.1395	0.6718	0.5684	Riccati
0	0.7725	0.8272	0.3900	0.2729	lin
0.1	0.8155	0.7717	0.8676	0.3452	lin
0.3	0.7713	0.7194	0.6124	0.3354	lin
1	0.7906	0.7230	0.7039	0.3498	lin
3	0.8265	0.8105	0.8676	0.3579	lin

The penalized objective is

$$J_w(\theta) = \text{cost}(\theta) + w \frac{\max(0, r_t - \hat{r}(\theta))^2}{r_t^2}, \quad (16)$$

a one-sided squared hinge—exceeding the target is never punished—normalized so that w is dimensionless and comparable to the score. The search remains restricted to the $u(0) = 0$ null space of §8.

9.3 The front, and a result in the opposite direction

The expected outcome was a strict trade. Instead, $w = 0.3$ **dominates** $w = 0$ **on all three axes**: better score (0.7713 against 0.7725), better held-out score (0.7194 against 0.8272), and a larger certified ball (0.3354 against 0.2729). The held-out column explains why. The unpenalized search overfits its six training initial conditions, degrading 7% out of sample, and the certificate penalty suppresses precisely the aggressive behaviour responsible.

A Lyapunov certificate acts as a regularizer. It charges for the same aggression that costs generalization, so the first increment of certificate-awareness is free.

Past that point the trade is real and w prices it: over $w \in \{0.3, 1, 3\}$ the certified ball rises $0.3354 \rightarrow 0.3498 \rightarrow 0.3579$ while the score degrades $0.7713 \rightarrow 0.7906 \rightarrow 0.8265$.

10 Calibrating the proxy

§9 left the certificate a dial rather than a specification: the proxy over-stated the certified ball by 1.18 to $2.51\times$, so asking for 0.67 delivered 0.35 . We had attributed this to S-procedure conservatism, which would have required a higher multiplier degree. It was instead *estimator error*, of two distinct kinds.

10.1 Two defects, both in the estimator

A minimum-order statistic. Violations are not rare—37% of cloud points satisfy $\dot{V} \geq 0$ —so the sampled minimum in (15) depends entirely on whether some point happened to land near the inner boundary. Across five seeds it reports $0.6718, 0.6744, 0.7484, 0.6608, 0.7369$: a 12.5% spread. Fixing the seed made it deterministic, not accurate.

Table 7: Estimator comparison, same controller and same necessary condition.

estimator	seed 0	seed 1	seed 2	seed 3	seed 4	spread	vs SOS
cloud	0.6718	0.6744	0.7484	0.6608	0.7369	12.5%	1.182×
ray	0.5937	0.5934	0.5849	0.5937	0.5915	1.5%	1.044×

The obvious repair is degenerate. Sharpening (15) by locally solving $\min\{z^\top Pz : \dot{V}(z) \geq 0\}$ fails structurally rather than numerically: $z = 0$ **is feasible**, since $\dot{V}(0) = 0$ exactly. The program has global optimum 0 at the origin and a descent method walks to it. Measured: of six local solves from the best sampled starts, two collapsed to $\|z\| = 0$ and all six terminated infeasible on the $\dot{V} = 0$ boundary, so a strict feasibility recheck discards every one and the refined bound never moves.

10.2 Ray search

Both defects vanish if the search is organized by direction. Along a fixed unit vector d define the first violation radius

$$r(d) = \min\{r > 0 : \dot{V}(rd) \geq 0\}, \quad \hat{\rho} = \min_d r(d)^2 d^\top P d, \quad (17)$$

computed by scan-and-bisect over a fixed direction set. There is no optimizer, no degenerate solution, and the origin is excluded by construction because the scan starts at $r > 0$. Bisection returns the *outer* bracket endpoint, which is a state genuinely satisfying $\dot{V} \geq 0$, so the necessary-condition property that makes the proxy safe to optimize against is preserved exactly. This is checked rather than argued: all 973 returned endpoints satisfy $\dot{V} \geq 0$ on re-evaluation, and the binding one is the explicit state $z^* = (+0.0237, +0.0778, +0.0628, -0.2563)$ with $\dot{V} = +2.9 \times 10^{-9}$.

10.3 What remains is a constant

Perturbing the fit along the $u(0) = 0$ null space produces controllers spanning a range of aggressiveness, each still with an equilibrium at the origin. Over five that certify, the ratio of the true certificate to the ray estimate is

$$\kappa = \text{median}\left(\frac{\text{SOS}}{\text{ray}}\right) = 0.9620, \quad \text{range } [0.9271, 0.9735]. \quad (18)$$

Of the 3.8% shortfall, 2.5 points are the deliberate 0.95 shrink applied to ρ before re-certifying ($\sqrt{0.95} = 0.9747$). Genuine degree-4 S-procedure conservatism is therefore about 1.3%—confirming that chasing multiplier degree would have bought nothing.

10.4 A specified radius, confirmed by the solver

Hinging (16) on the calibrated estimate $\kappa\hat{r}$ states the target directly in certified-ball units. A pass that misses raises the target by exactly the shortfall; a pass that misses *and* fails to improve on the previous one is not short of target but stuck, and the target is pushed 50% further.

All three specifications are met. The best controller of the entire study falls out of this procedure: **certified ball 0.6791 at score 0.8469**, which dominates the imitation controller on *both* axes—a larger certified region than 0.5684 and a 19% better score than 1.0437—and nearly doubles the best certified region obtained in §9.

Table 8: Specifying a certified radius. Every “yes” is a semidefinite-program certificate, not a proxy estimate.

spec	pass	score	held out	$\kappa\hat{r}$	SOS Ric	SOS lin	best	met?
0.35	1	0.7956	0.7546	0.3478	0.0000	0.3415	0.3415	no
0.35	2	0.7962	0.7227	0.3577	0.0000	0.3549	0.3549	yes
0.45	1	0.7690	0.7296	0.5277	0.0000	0.3452	0.3452	no
0.45	2	0.7857	0.7536	0.5865	0.0000	0.3430	0.3430	no
0.45	3	0.8487	0.8667	0.7269	0.6782	0.3654	0.6782	yes
0.55	1	0.7717	0.7466	0.5503	0.1038	0.3368	0.3368	no
0.55	2	0.8469	0.8560	0.7278	0.6791	0.0000	0.6791	yes

10.5 Two limits calibration does not remove

A single κ can fail. At specification 0.45, pass 1, $\kappa\hat{r} = 0.5277$ against a certificate of 0.3452: a ratio of 0.65, far outside $[0.927, 0.974]$. The per-candidate columns show why—SOS Ric = 0.0000. The proxy takes a maximum over Lyapunov candidates and attains it at the Riccati matrix, where $\dot{V} < 0$ genuinely does hold that far out, but which the degree-4 S-procedure cannot certify *at all*. The proxy measures where $\dot{V} < 0$ *is true*; SOS measures where it is *provable at this degree*; the argmax of the two differs. Calibration handles the gap when both agree on the candidate, and the outer loop absorbs it when they do not.

The achievable set is bimodal. Every row lands near either 0.35 or 0.68, according to which Lyapunov function certifies—two basins, not a continuum. Specification 0.45 therefore fails twice and then succeeds, and asking for 0.45 delivers 0.6782: the specification is *met, not matched*. This is a property of the landscape, not of the calibration. Both limits point at the same remedy: searching V jointly with θ , a bilinear problem, would remove the max-over-two-candidates artifact that produces the calibration outlier and the bimodality alike.

11 What is proved, and what is not

Table 9 separates the results by evidentiary status: analytically proven, certified numerically, or unavailable. We regard keeping this distinction visible as part of the method rather than as book-keeping, because the value of a variational formulation lies precisely in which claims it can discharge without a solver.

Two obstructions did not yield and should be stated plainly. G13 is genuinely open: the antecedent landscape is multimodal and nothing here bounds the gap to the global optimum. G14 is not an open problem but a structural exclusion—a supremum and an exit time are not integral functionals, so no argument of the form of Theorem 14 can reach them. The honest route to certifying the benchmark objectives is to *restate* them as integral costs ($\int \phi(u)dt$ with ϕ a convex barrier; $\int w(t)\|x\|^2 dt$), which changes the problem in a direction the framework can certify.

12 Machine verification

Three tiers of checking were applied, weakest to strongest.

Tier 1: symbolic identity checking. Twenty-one identities—the Euler–Lagrange equation, Galerkin orthogonality, the envelope-theorem gradient, the λ^* integrals, the Bregman identity, the

Table 9: Guarantee inventory.

#	Guarantee	Status	Conditions
G1	Consequents globally optimal, closed form	Proven	antecedents fixed; independent regressors
G2	Action convex in function space	Proven	$\lambda \geq 0$
G3	Reduced gradient exact	Proven	δ -coverage
G4	Greedy = joint iff Gram block-diagonal	Proven	for all targets
G5	$\lambda^* = 3b^4/(8c^2)$	Proven	2 equal-width Gaussians
G6	$\vee = +$ iff disjoint output supports	Proven	—
G7	Centroid = order-0 TSK	Proven	disjoint, product implication
G8	Annealing error bound	Proven	—
G9	TSK reproduces LQR exactly	Proven	LTI, quadratic cost (and vacuous)
G10	Bregman suboptimality identity	Proven	control-affine, convex c , admissible, V^* known
G11	Local optimality of antecedents	Certified numerically	KKT on critical cone + FD Hessian
G12	Stability / region of attraction	Proven by SOS	rational MFs, $u(0) = 0$, interior-point SDP
G13	Global optimality over antecedents	Not available	—
G14	Certificate for settling time, peak force	Provably outside G10	not integral functionals

partition-of-unity consequences—are verified as symbolic identities in a computer algebra system rather than at sampled points. All 21 pass. (One subtlety: a matrix comparison `Matrix(...) == 0` is `False` even for the zero matrix; using the correct predicate moved the count from 19/21 to 21/21.)

Tier 2: exact rational proof of the SOS certificate, as described in §8: residual exactly zero, all 14 integer leading principal minors positive.

Tier 3: proof assistant. A Lean/Mathlib formalization would close the gap Tier 1 leaves—the analytic hypotheses—but could not be attempted here, as the toolchain host is unreachable under this environment’s network policy. Rather than commit unverifiable source we record where it would pay. Theorem 14 is the clear candidate: its content is an integration-by-parts argument plus a limit, and its hypotheses (admissibility, convexity, differentiability of V^*) are exactly the kind that get quietly dropped—as §6 notes, we dropped one empirically. The SOS side should stay in exact integer arithmetic, where it already is: formalizing it would mean formalizing Bareiss determinants to re-derive an answer that is already decidable and decided.

13 Limitations and future work

- **One plant, one dimension of universe.** The approximation results are scalar-input; λ^* in the multi-input case turns the crossover into a hypersurface with a direction-dependent width and may have no clean form. The control results use a single benchmark.
- **Certification does not scale.** At $N = 8$ the SDP is 715×715 and out of reach with these solvers. This bounds the whole certified story to low rule count.
- **Two fixed Lyapunov candidates.** Every certified radius reported is a maximum over the Riccati matrix and the closed-loop linearization. Searching V jointly with θ is bilinear, was not attempted, and is the single change that would most improve §10.
- **The classifier results have not met real data.** Theorems 10–12 are proved but exercised only on synthetic partitions. This does not test the guarantees, which are already proven; it

tests whether they bind on anything real.

- **Relative form of Theorem 14.** The certificate presupposes V^* , which is what a forward problem lacks. For any $\hat{V}$ satisfying HJB with residual ε , the same algebra should bound the true gap in terms of ε ; the useful part of that derivation is determining in which norm ε must be measured. Without it, the certificate is restricted to inverse-optimal benchmarks—problems whose answer was known in advance.
- **Analytic Hessian.** G11 is certified numerically only because the Hessian is obtained by finite differences. The same envelope argument differentiated once more would upgrade it.

Reproducibility

All results are produced by a deterministic, seeded implementation; recorded output is stored alongside the source, and the numerical constants that matter (quadrature order, ridge scaling, optimizer schedule, finite-difference steps, definiteness tolerances) are documented individually together with the failure each was chosen to prevent. Tables 3, 4, 6, 7 and 8 are transcribed directly from that recorded output.